PRANAY PANDEY

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EXPERIENCE

Adobe – *Product SWE Intern* May 2026–Jul 2026

Tech: Python, Scala, AWS-Bedrock Anthropic Models, Azure OpenAI Services, Langfuse, Jenkins CI/CD, Git, Jira

- Designed a GEPA-inspired, **feedback-based prompt optimization loop** for the MAS LIPI agent, integrating deterministic evaluation gates and LLM-driven candidate generation, elevating the system pass rate from **76% to 87%** across 628 test scenarios with **zero runtime overhead**.
- Configured LLM telemetry via Langfuse and Jenkins CI/CD pipelines. Benchmarked GPT-5.2 against different frontier models. Proved GPT-5.4-mini (low reasoning) **achieved a 57.5% operational cost reduction** while sustaining a 97.0% deterministic-set pass rate.
- Developed **Meta Harness (MHSC), an agent-agnostic harness optimization framework** inspired by Stanford's Meta-Harness. Automated prompting guide-driven edits and iterative benchmarking, **reducing ~3 weeks of manual LLM model migration to an overnight run** across Adobe's MAS Agents (Halo, LIPI, Rules). MHSC reduced per-scenario token consumption by **12.33% in LIPI** and **27.0% in Halo**.

Defence Research and Development Organisation (DRDO) – *Machine Learning Intern* Jun 2025–Jul 2025

Tech: Python, Pandas, NumPy, RapidFuzz, Scikit-learn, Node2Vec, Fuzzy C-Means, NLTK, FastAPI, Docker, Pytest

- Built a scalable, **hybrid candidate-matching microservice** utilizing Fuzzy C-Means, Node2Vec embeddings, and TF-IDF; slashed matrix generation from **~1hr to 9.79s** while maintaining **83.6% Top-1 accuracy** by integrating C++ RapidFuzz and resolving N-gram boundaries.
- Implemented a **fully vectorized NumPy scoring pipeline** processing 200,000 multi-dimensional matches in **0.90s (~325,000 pairs/sec)**; hardened the system using $O(1)$ **disk-caching** for near-zero latency warm boots and a secure, rootless Docker containerization pipeline.

EDUCATION

2023–2027	**B.Tech (Computer Science Engineering)**	Delhi Technological University (Formerly DCE)
2021–2023	**CBSE (Class XII)**	Sant Gyaneshwar Model School, Delhi
2009–2021	**CBSE (Class X)**	Hansraj Model School, Punjabi Bagh, Delhi

ACADEMIC PROJECTS

RAG-Driven Document Q&A and Smart Chat-Bot Platform (GitHub Link)

Tech: Python, LlamaIndex, BM25, CrossEncoders, Groq API (LLaMA-4), Hugging Face, FastAPI, Node.js, Express, MongoDB

- Built a Python RAG microservice, implementing a **2-stage hybrid retrieval pipeline (Dense + BM25)** fused via Reciprocal Rank Fusion (RRF). Integrated a HuggingFace CrossEncoder to jointly score query-passage pairs, surfacing the top-3 most **contextually relevant** chunks.
- Accelerated throughput by developing a cosine-similarity **semantic caching** layer over query embeddings, reducing response latency by **98.3% (from 3.55s to 60ms)**. Configured a deterministic 13-class intent router to bypass generative LLM inference for static interactions.

Modular C++ Ride-Sharing System — Spatial Matching Engine (GitHub Link)

Tech: C++, Drogon Web Framework, Domain-Driven Design (DDD), RESTful APIs, Atomics, Object Pool, Spatial Grid

- Designed a C++ backend utilizing 4-layer DDD with decoupled pricing and matching engines. Implemented a custom freelist Object Pool over contiguous `aligned_storage_t`, achieving **27ns** allocation latency by bypassing kernel overhead for **1,000+** GPS telemetry events/sec.
- Developed a high-performance Drogon REST API featuring a cache-friendly Spatial Grid that replaced linear $O(n)$ scans with $O(1)$ lookups, achieving **26µs matching latency (a 17.4x speedup)** for 10,000 drivers. Integrated lock-free surge pricing, ensuring zero heap allocations.

AEGIS: Local-First LLM Agent Optimization & Trajectory Prediction Gateway (GitHub Link)

Tech: Python, FastAPI, Redis, SQLite, Sentence-Transformers, NetworkX, Node2Vec, Ollama, vLLM

- Created a **zero-runtime-LLM** gateway delivering exact-match and verifier-gated **semantic caching**, heuristic cascade routing, and shadow-mode policy staging, powered entirely by dense local embeddings, graph-based ANN retrieval, and zero-cost inference via **Ollama & vLLM**.
- Integrated **Trajectory-Based Prompt Enrichment** with fail-closed safety gates against adversarial injection replay and cross-tenant data leakage, and deployed a **two-plane evaluation suite** utilizing a **CPU judge** to statistically validate metrics.

TECHNICAL SKILLS

- **Languages:** Python, C++, C, Java, Rust, JavaScript, Bash
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, Transformers, Hugging Face, LlamaIndex, LangChain, LangGraph, OpenCV, NumPy, Pandas, GenAI
- **Backend, Databases & Distributed Systems:** REST APIs, FastAPI, Node.js, Express, React, Vercel, Next.js, GraphQL, Microservices, Kafka, RabbitMQ, Amazon SQS, Redis, NGINX, Amazon S3, PostgreSQL, MySQL, MongoDB, SQLite, Neo4j, NoSQL
- **Cloud & Infra:** AWS (Amazon Web Services), Docker, Kubernetes, Jenkins, Git, GitHub Actions, CI/CD, Prometheus, Grafana, Langfuse, Linux
- **Computer Science Fundamentals:** System Design (HLD, LLD), Operating Systems, Computer Architecture & Organization, Compiler Design, Parallel Computing, Computer Networks, Database Management, Object-Oriented Programming, Data Structures

ACHIEVEMENTS AND CERTIFICATIONS

- **340+ LeetCode** Data Structures and Algorithms Problems – Python and Java (LeetCode Profile)
- **2nd Place, Adobe Invictus DevCraft Hackathon** – Programmed a high-precision Naive Bayes Document Classifier (95%+ accuracy)
- **Machine Learning Specialization** (DeepLearning.AI & Stanford University) – Supervised Machine Learning, Unsupervised Learning, Reinforcement Learning and Advanced Learning Algorithms (Regression, Clustering, Ensembles and Recommender Systems) – (Certificate Link)
- **Deep Learning Specialization** (DeepLearning.AI) – Neural Networks and Deep Learning, Sequence Models, Convolutional Neural Networks (Natural Language Processing, Transfer Learning, Transformers, RNNs, LSTMs and Hugging Face) – Certificate Links: (CL1) (CL2) (CL3)
- **Generative Adversarial Networks (GANs) Specialization** (DeepLearning.AI) – (Certificate Link)
- **Google Cloud Platform – Courses:** Cloud AI Infrastructure, Agentic Architectures & Cloud Cybersecurity (Skills Profile)